

GIFT: Glove-Inferred Force Transfer

Force-Aware Human-to-Robot Skill Transfer from a Wearable Sensing Glove to a Robot Hand Without Tactile Sensors

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Abstract—Human-to-robot skill transfer from sensing gloves has so far relied on shared hardware: the same tactile glove worn by the demonstrator and the robot, or a learned alignment between two tactile sensors. We present GIFT (Glove-Inferred Force Transfer), a pipeline in which the interface between human and robot is a physical unit rather than a shared sensor: fingertip force is measured in newtons on the human side and estimated in newtons on the robot side. A wearable glove records finger flexion, calibrated fingertip force, and wrist orientation, while a head-mounted camera records the demonstration; no robot is present during data collection. At deployment, the robot estimates force from actuator-current residuals relative to a free-space baseline, through a calibrated mapping to newtons, so any position-controlled hand that reports motor current can serve as the deployment platform. The robot hand carries no tactile sensors. The policy uses a glove-space state representation and predicts finger-position targets; the robot enters only through two calibrated adapters, a retargeting decoder and a force estimator. We evaluate GIFT on a cup grasp-and-hold task with two action-chunking policies trained on the same demonstrations, with fingertip-force inputs retained in one and zeroed in the other. In a 50-rollout evaluation with sample size and metrics fixed before scoring, both policies succeeded in all 25 rollouts. The median of the per-rollout hold-phase grip-force estimates was 53% lower with force inputs: 1.20 N versus 2.55 N (one-sided Mann–Whitney U, $p < 0.0001$). In an observation ablation, a vision-only policy achieved 0/15 grasps, policies given hand-command state acquired the grasp, and the force inputs determined how hard the policy held. A force channel measured on the human hand thus transfers to a robot hand with no tactile hardware, through a retargeting map from five glove channels to seven robot actuators, with no sensor shared between the two.

Index Terms—Tactile sensing, learning from demonstration, dexterous manipulation, force control, human-to-robot skill transfer.

I. INTRODUCTION

Contact-rich manipulation is regulated by touch. A person holding a thin plastic cup applies just enough grip not to drop it and little enough not to crush it, and does so without looking, because the controlling signal is force rather than appearance. Imitation-learning pipelines built on cameras alone must infer this hidden variable from pixels, and on tasks where the visual scene barely changes with grip force, they cannot.

The value of tactile signals in human demonstrations is now well established: tactile gloves transfer human demonstrations to robots [1, 2], and cross-sensor alignment transfers tactile policies between sensing hardware [3]. These pipelines close the human-to-robot gap by sharing hardware or by learning to bridge it. OSMO puts the same tactile glove on the demonstrator

GIFT sensing glove



- ✓ Fingertip force in newtons
- ✓ Finger curl and wrist orientation
- ✓ Head camera, no robot present
- ✓ Deploys on a hand with no tactile sensors

Human-to-robot skill transfer

Human glove demos



glove FSRs report N

force as a
physical unit (N)
no robot data

Robot policy



no tactile sensors;
actuator currents estimate N

Fig. 1. **GIFT**. Top: the sensing glove (fingertip sensors highlighted) captures fingertip force in newtons, finger curl and wrist orientation, alongside a head camera; no robot is present during collection. Bottom: the policy trained on these demonstrations is deployed on a robot hand with no tactile sensors, whose force observation is estimated in newtons from its own actuator currents. The human and the robot share a physical unit, not a sensor.

and the robot, so that the glove itself is the interface [1]; TactAlign learns an alignment between the human’s and the robot’s tactile sensors [3]. In both cases the robot carries tactile sensing at deployment.

GIFT takes a different route: the interface between human and robot is a physical unit. Force is recorded in newtons on the glove and estimated in newtons on the robot, so the two sides never share a sensor, and the robot needs none. This rests on two design decisions. First, force is captured as a physical quantity: the glove’s fingertip force-sensitive resistors (FSRs) are calibrated to newtons against a reference scale, so the recorded channel is not tied to any particular sensor. Second, force at deployment is *estimated* rather than sensed: following the residual principle of force estimation from actuation [4], the robot hand’s actuator currents are compared against a free-space baseline and mapped into the same newton range the policy saw in training. The policy itself observes and acts in glove space; the robot appears only through two calibrated

adapters, a retargeting decoder from glove space to actuator commands and the force estimator. Nothing in the pipeline is specific to the hand used here; any position-controlled hand that reports motor current provides the deployment-side signal. The capture rig is a glove with five flex sensors, five fingertip FSRs, and a wrist IMU, read by an ESP32-S3 microcontroller, together with a head-mounted camera. No robot is present during data collection (Fig. 1).

We evaluate GIFT with two action-chunking policies [5] trained on the same demonstrations of a cup grasp-and-hold task, with identical architecture, hyperparameters, and seed, differing in exactly one thing: the fingertip-force inputs are retained in one policy and zeroed in the other. The policies are deployed on a seven-actuator tendon-driven hand with no tactile sensing. Both policies succeeded in all 25 of their rollouts, but the policy with force inputs held the cup at less than half the estimated grip force. A third policy trained on vision alone never acquires the grasp, which places the contribution of each observation channel on a single ladder.

Contributions.

- 1) **GIFT, an end-to-end pipeline for human-to-robot skill transfer of force without tactile hardware on the robot:** a calibrated sensing glove and head camera for capture, a policy with a glove-space state representation that predicts finger-position targets, spline-based retargeting from glove space to hand actuators at runtime, and current-residual force estimation at deployment. The pipeline requires no robot during capture and no tactile sensor at deployment, and the human and robot share no sensor.
- 2) **A 50-rollout result on real hardware:** at saturated task success (25/25 for both policies), the policy with force inputs held at **53% lower estimated grip force** (median of per-rollout hold-phase medians, 1.20 N versus 2.55 N, one-sided Mann–Whitney U, $p < 0.0001$).
- 3) **An observation ablation** isolating the channels: a vision-only policy fails to grasp in 0/15 rollouts against 3/3 for in-session controls (Fisher exact $p \approx 0.0012$); hand-command state acquires the grasp; the fingertip-force inputs determine how hard the policy holds.

II. RELATED WORK

Tactile gloves for human-to-robot skill transfer. OSMO [1] is the closest antecedent: an open-source magnetic tactile glove worn by both the human demonstrator and the robot hand, so that the glove itself is the shared interface and the visual and tactile embodiment gaps are minimized. Feel the Force [2] collects contact demonstrations with an AnySkin-augmented glove [6] and reports higher success than policies trained from teleoperation. GIFT differs in two respects: the capture hardware is piezoresistive (FSRs and flex sensors rather than magnetic skins), and the robot carries no tactile sensor at all, so the deployment-side signal is estimated from actuation rather than sensed. The human and the robot share a unit, not a sensor.

Robot-free demonstration capture. Handheld and wearable interfaces decouple demonstration collection from robot access: UMI’s handheld gripper [7], exoskeleton and wearable dexterous-hand interfaces [8–10], and in-the-wild egocentric pipelines [11, 12] all scale collection by removing the robot from the loop. These systems predominantly capture kinematics and vision. GIFT adds calibrated fingertip force as a first-class captured signal, which our ablation shows to be the channel that determines grip behavior.

Tactile sensing and representations for manipulation policies. Dense tactile arrays improve fine-grained manipulation policies [13, 14]; vision-language-action models augmented with tactile inputs gain force-aware behavior [15]; and representation learning transfers across heterogeneous tactile sensors [16, 17]. A complementary line removes deployment-time tactile hardware by *imagining* tactile signals from vision [18]. GIFT shares the goal of tactile-free deployment but takes the actuation route: the hand’s own motor currents, baselined and calibrated, stand in for the missing sensor [4, 19].

III. THE GIFT PIPELINE

GIFT is built around four design principles. (1) *Force is a physical unit:* the captured channel is expressed in newtons, so any instrument that produces newtons can stand in for the glove at deployment. (2) *No robot during capture:* demonstrations are recorded on the human hand with a head camera, so collection scales with people rather than with robots. (3) *No tactile sensor at deployment:* the robot’s own actuator currents, baselined in free space, provide the force observation. (4) *One instrument scores both policies:* the with-force and without-force policies are compared with the same deployment-side estimate, so the force channel is the only difference between them.

The pipeline has four stages: capture on the human with a sensing glove and head camera (§III-A, §III-B), retargeting from glove space to robot actuators (§III-C), policy training on the demonstrations (§III-D), and deployment with force estimated from actuator current (§III-E).

A. Sensing glove

The glove mounts three sensor groups (Fig. 2). Five flex sensors, one per finger along the dorsal side, measure finger curl. Five force-sensitive resistors at the fingertips measure normal contact force. A WitMotion IMU at the wrist provides wrist orientation (roll, pitch, yaw). All channels are read by a Seeed XIAO ESP32-S3 microcontroller and streamed over USB serial to the recording host. Fingertip force is calibrated to newtons against a digital reference scale with a fitted curve (Fig. 3A), so that the recorded force channel is a physical quantity rather than a raw sensor reading. This is the property that lets the deployment side substitute a different instrument (§III-E).

B. Demonstration capture

A head-mounted camera records the egocentric view. The demonstrator wears the glove and performs the task with their own hand; no robot is present or required. All streams share a



Fig. 2. **Hardware.** (A, B) The sensing glove, palm and dorsal sides, with sensor locations highlighted: five fingertip force-sensitive resistors (discs) and five flex sensors along the fingers (streaks); the wrist enclosure holds the ESP32-S3, the IMU and the LED/buzzer start cue. (C) Deployment setup (illustration): the TetherIA Aero Hand Open in a plain nitrile glove with no sensors, mounted on an SO-101 arm, the cup at its marked position, and the camera on a fixed mount aligned to the demonstrator’s viewpoint. (D) Capture signal path: all glove channels and the camera share one time axis on the recording host.

single time axis on the recording host, and an audible start cue captured in both the sensor log and the video verifies alignment. The demonstration corpus for this work is 73 sessions and 38,150 frames of a constant-force cup grasp-and-hold task.

C. Retargeting from glove space to robot actuators

The target platform is a TetherIA Aero Hand Open (seven actuators, sixteen joints). The policy acts in glove space: its actions are five finger targets in the glove’s flex coordinates, and a per-finger mapping decodes them into actuator commands at runtime, so retargeting is an adapter around the policy rather than a transformation of the training data. The thumb, whose single flex channel drives three joints, uses monotone PCHIP splines fitted through captured calibration anchors (open pose, intermediate apertures, and the task hold pose); thumb opposition rides the spline, so as the hand closes the thumb sweeps into opposition as the demonstrator’s did. The hold-pose anchor is pinned to the median demonstrated hold command, so the policy’s hold action decodes to the demonstrated grip geometry.

D. Policy

We train action-chunking transformer (ACT) policies [5] with LeRobot [20]. The observation is the egocentric image plus a 13-dimensional state vector: five flex, five fingertip forces, and three IMU angles. The action is the five finger targets in glove flex coordinates (§III-C). To isolate the force channel we train two policies on the same corpus. The **with-force** policy sees all 13 state dimensions; the **without-force** policy has the five force dimensions zeroed at training time. Architecture, preprocessing (a uniform FSR baseline shift and pinch-symmetry imputation), image augmentation, 20k training steps, batch size, and seed

are identical. At deployment both policies receive identical inputs except the force channels (robot-side force estimates versus zeros, exactly as trained).

E. Deployment-time force from actuator current

The robot hand has no tactile sensors. Following the residual principle of force estimation from actuation [4], we log a free-space baseline of actuator current versus hand position (no object) and, at deployment, compute per-actuator residuals

$$i_{\text{contact}} = i_{\text{measured}} - \hat{i}_{\text{free}}(q). \quad (1)$$

A tendon-driven hand performing a static hold permits this position-indexed baseline where a full arm would require a learned dynamics model (Fig. 3B, C). Residuals map to newtons through a fitted calibration and are normalized into the same range as the training-set glove FSRs, keeping the policy’s force observation in distribution. The policy’s state therefore stays in glove space at deployment as well: the flex channels are the policy’s own commanded glove-space pose fed back as state (under position control the hand tracks the command until contact), the IMU channels are held at values representative of the demonstrations since the wrist trajectory is fixed, and the force channels carry the current-based estimate in newtons. The robot enters the loop only through the retargeting decoder and this estimator. For reporting, force values in Fig. 4 and Table I are recomputed from the logged raw signal using a digital-scale calibration of the robot fingertips; this reporting calibration was declared before scoring, and runtime behavior used the frozen deployment estimator throughout.

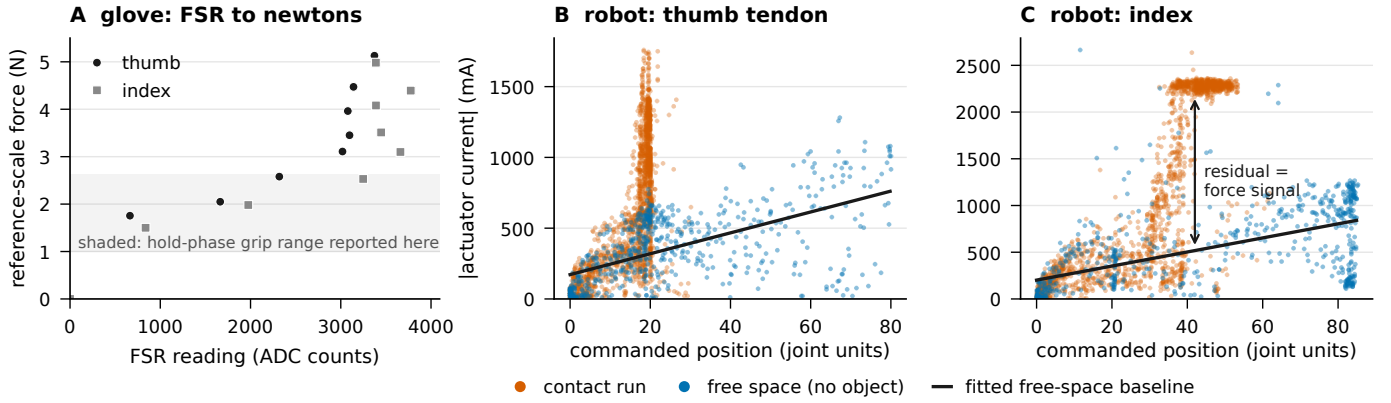


Fig. 3. **Two instruments, one unit.** (A) Glove side: fingertip FSR readings against a digital reference scale; the shaded band is the hold-phase grip range reported in §IV-D. (B, C) Robot side: actuator current against commanded position for the thumb-tendon and index actuators, from the free-space log (no object) with its fitted linear baseline, and from a contact log. At the grasp position the contact current rises vertically above the baseline; that residual, mapped to newtons, is the policy’s deployment-time force observation.

IV. EXPERIMENTS

A. Setup

Platform. The Aero Hand is mounted on an SO-101 arm executing a pre-programmed wrist-only trajectory (approach the cup, grasp window, lift, settle). The lift is triggered when the deployment force estimate reports a sustained grip (> 0.5 N for 1 s) or after an 18 s timeout, with the same rule for both policies. The policy controls the hand only, under position control with identical torque caps for both policies; the action chunk length matches training ($n_{\text{action_steps}} = 100$).

Task and object. Grasp-and-hold of a plastic cup of the same type used in the demonstrations, placed empty at a marked position. The arm trajectory delivers the hand; the policy performs the grasp and hold. The cup returns to the mark between rollouts.

B. Protocol

Sample size, metrics, thresholds, and the headline decision rule were fixed before the first scored rollout, and all completed rollouts are reported. The sample is $n = 50$ rollouts, 25 per policy, in strict with/without alternation with no extensions. Per-rollout success is defined as grasped and held intact through the run, recorded by the operator as grasped, crushed, dropped, or held, with video retained. The per-rollout grip-force metric is the median, over hold-phase contact frames (> 0.3 N), of the mean thumb-index force estimate in newtons, compared across policies by a one-sided Mann-Whitney U test (with-force $<$ without-force, $\alpha = 0.05$) on successful rollouts. The pre-declared decision rule selects the headline: if success differs (Fisher exact $p < 0.05$, at least 25 percentage points), the headline is reliability; if success saturates or does not differ but grip force does, the headline is force regulation with success reported as no difference. Both outcomes are reported regardless. Camera alignment and arm pose were probed every eight rollouts, and exclusion rules for truncated logs and stale frames were declared in advance.

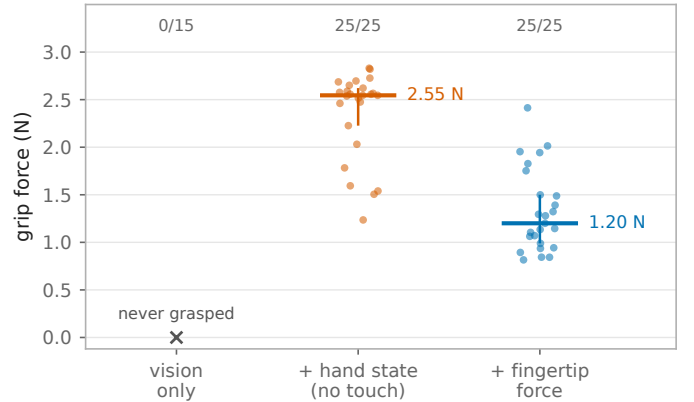


Fig. 4. **Observation ablation on the cup grasp-and-hold task.** Tallies along the top are grasp success. The vision-only policy never acquires the grasp (0/15; interleaved controls in the same session succeeded 3/3), so it has no grip-force distribution. Adding hand-command state acquires the grasp, and adding the fingertip-force channel does not change whether the policy holds (25/25 for both) but changes how hard: median grip force falls from 2.55 N to 1.20 N (-53% , one-sided Mann-Whitney $p < 0.0001$). Each marker is one rollout ($n = 25$ per policy); bars show median and interquartile range. Force is a current-based estimate calibrated against a reference scale (§III-E). The vision-only arm was evaluated in a separate lamp-matched session with interleaved controls; force comparisons are same-session only.

C. Baselines

Three policies form the observation ladder. **Vision only:** the entire state vector zeroed at training and deployment. **Vision + hand state:** flex and IMU channels present, force zeroed (the without-force policy). **Vision + hand state + fingertip force:** the full observation (the with-force policy). All three share identical demonstrations, images, actions, architecture, training steps, and seed.

D. Results

Same success, half the grip force. All 50 rollouts completed with no exclusions. Task success saturated at **25/25 for both policies**, so the decision rule selects force regulation as the

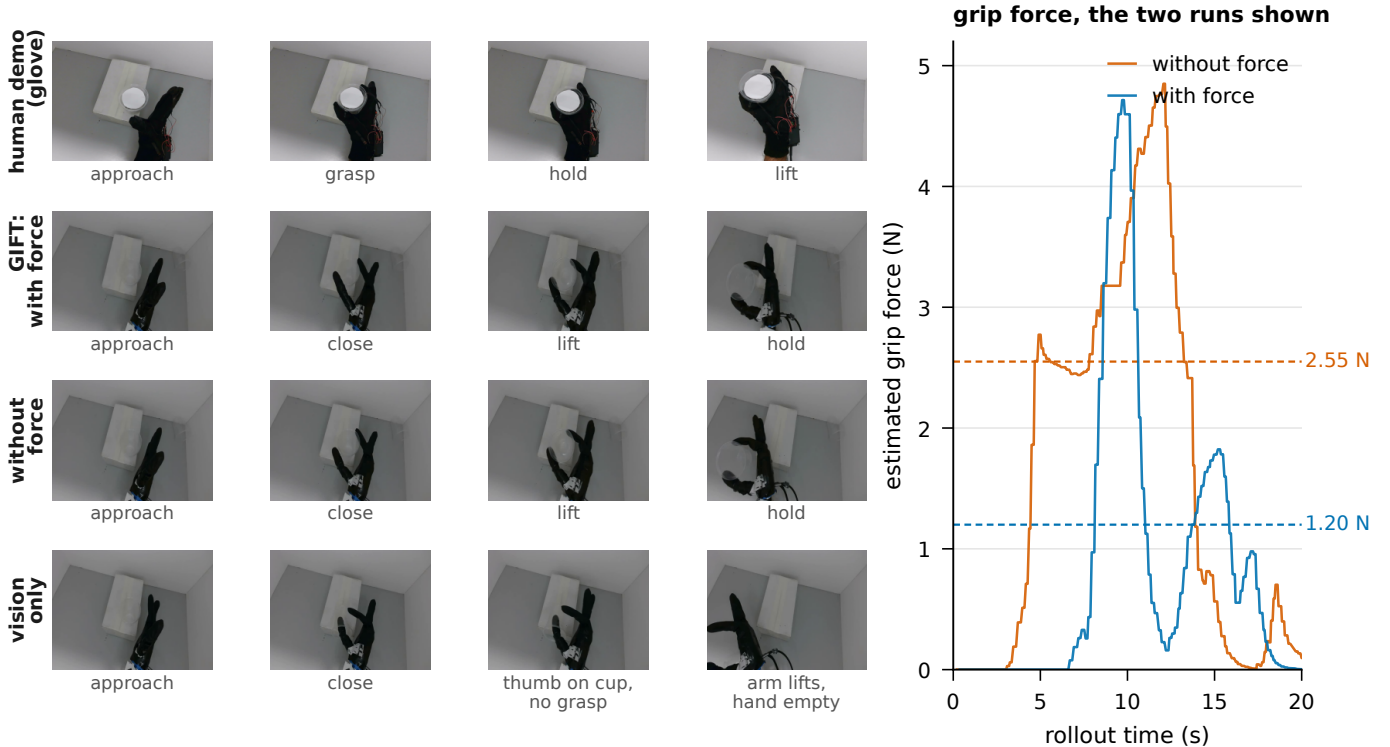


Fig. 5. **Representative rollouts from the head camera.** Top row: a human demonstration with the sensing glove. Middle rows: the with-force and without-force policies on the robot hand, showing for each arm the run whose hold-phase median is closest to its arm’s median (1.20 N and 2.55 N). Bottom row: the vision-only policy lands the thumb on the cup, bottoms out the close without a grasp, and the arm lifts an empty hand. Right: the logged grip-force estimate over the two state-policy runs shown, with their hold-phase medians dashed.

TABLE I
OBSERVATION ABLATION ON THE CUP GRASP-AND-HOLD TASK. HAND-COMMAND STATE ACQUIRES THE GRASP; THE FINGERTIP-FORCE CHANNEL CHANGES HOW HARD THE POLICY HOLDS, NOT WHETHER IT HOLDS. GRIP FORCE IS THE MEDIAN OF PER-ROLLOUT HOLD-PHASE MEDIAN ($n = 25$ PER POLICY). ALL COMPLETED ROLLOUTS REPORTED.

Observations	Grasp	Hold	Grip force (N)
Vision only	0/15	—	—
+ hand state (no touch)	25/25	25/25	2.55
+ fingertip force	25/25	25/25	1.20

headline. The median of the per-rollout hold-phase grip-force estimates was **1.20 N** (IQR 0.99–1.50) with force inputs versus **2.55 N** (IQR 2.23–2.62) without, a **53% reduction** (one-sided Mann–Whitney U, $z = 5.34$, $p < 0.0001$; Fig. 4; representative rollouts in Fig. 5). A drift check found no session trend ($\text{corr}(\text{run}, \text{median}) = 0.01$). Grasp-transient peaks were statistically indistinguishable across policies and sat near the estimator’s ceiling, so they are not quoted as measurements. The without-force policy does not fail the task; it succeeds by squeezing more than twice as hard as the policy with force inputs, because nothing in its observation indicates when the grip is sufficient. We do not compare either policy against the demonstrator’s own grip force: the glove FSRs and the deployment-side estimator are different instruments, and only the policy-versus-policy comparison holds the instrument fixed.

Vision alone does not grasp. The vision-only policy failed to acquire the grasp in **0 of 15** rollouts, while interleaved same-session control rollouts of the state-carrying policies went **3/3** (Fisher exact $p \approx 0.0012$; including two mislabeled but otherwise valid runs, 0/16 versus 4/4, $p \approx 0.0002$). The failure signature was consistent (Fig. 5, bottom row): the thumb landed on the cup and the commanded close bottomed out without a grasp. State-carrying policies grasped in every lighting condition tested across three sessions; the vision-only policy failed in both sessions in which it was tested. On this task the visual scene barely changes with hand configuration, so hand-command state is what carries the timing of the grasp, and the fingertip-force inputs are what regulate its strength (Table I).

Mechanism. The action space is five finger positions; force enters the policy as an observation and appears in no term of the loss. The 53% reduction is therefore force-informed imitation: observing force shifts which demonstrated positions the policy reproduces. This locates the next step directly, since the force-annotated demonstrations GIFT produces already support making force a prediction target without new hardware.

V. LIMITATIONS

Sensing hardware. The glove’s FSRs measure normal force only; shear, which matters for slip and for in-hand reorientation, is not captured. FSRs also exhibit drift and hysteresis, and the per-finger calibration must be repeated when a sensor is

replaced. On the robot side, the current-residual estimator is a relative instrument: it is identical for both policies, which is what the comparison requires, but absolute newton values are approximate, and the estimate saturates near 5 N, censoring the grasp transient. A load-cell reference object shared by glove and robot would put both instruments on one scale and enable direct human-to-robot force comparison.

Task scope. All results are on one object at one position with a fixed arm trajectory; the policy controls the hand only. Success saturated at 25/25 for both policies, so this evaluation measures force regulation, not reliability. Success was recorded by a single operator from a fixed alternation order, with video retained; the evaluation is not blinded. Extending to object variation, arm control, and tasks where over-gripping causes visible failure (crushing, slipping) is the natural next evaluation.

VI. CONCLUSION

We presented GIFT, a pipeline for human-to-robot skill transfer in which the interface between human and robot is a physical unit: calibrated fingertip force captured on a human hand with a wearable sensing glove is transferred to a robot hand with no tactile sensors, using actuator-current residuals as the deployment-side force estimate. On a cup grasp-and-hold task, two policies trained on the same demonstrations reach the same success, and the policy with force inputs holds at half the estimated grip force. The observation ablation decomposes the channels cleanly: vision alone does not grasp, hand-command state acquires the grasp, and the fingertip-force inputs turn a successful squeeze into a controlled hold. Because the policy uses a glove-space state representation, force is captured as a physical unit, and the deployment-side estimate comes from a signal every position-controlled hand provides, the pipeline carries no dependency on the sensing or actuation hardware used here.

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